

torch.hash_tensor#

https://pytorch.org/docs/stable/generated/torch.hash_tensor.html#torch.hash_tensor

Description:

Returns a hash of all elements in the input tensor. Currently only mode=0 (reduction via xor) is supported. The output will always be of type torch.uint64. The elements of input are upcasted to their 64 bit float / integer equivalent and bitcasted to torch.uint64 before reduction via xor.

input (Tensor) – the input tensor. mode (int) – The hash to use. Default: 0 (xor_reduction)

Function Signature

```
torch.hash_tensor(input, *, mode=0) → Tensor#
```

Parameters

Keyword Arguments

```
torch.hash_tensor(input, dim, *, keepdim=False, mode=0) → Tensor
```

Parameters

Parameters

Keyword

mode (int) – The hash to use. Default: 0 (xor_reduction)

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mode (int) – The hash to use. Default: 0 (xor_reduction)

Code Examples

```
>>> a = torch.randn(1, 3)
>>> a
tensor([[ 1.1918, -1.1813,  0.3373]])
>>> torch.hash_tensor(a)
tensor(13822780554648485888, dtype=torch.uint64)
```

```
>>> a = torch.randn(2, 4)
>>> a
tensor([[ 0.1317, -0.5554, -1.4724, -1.1391],
        [ 0.0778, -0.6070,  0.6375,  0.1798]])
>>> torch.hash_tensor(a, 1)
tensor([9233691267014066176, 9255993250844508160],
dtype=torch.uint64)
```

Detailed Documentation

Documentation

Returns a hash of all elements in the input tensor.

Currently only `mode=0` (reduction via xor) is supported. The output will always be of type `torch.uint64`. The elements of input are upcasted to their 64 bit float / integer equivalent and bitcasted to `torch.uint64` before reduction via xor.

`input (Tensor)` – the input tensor.

`mode (int)` – The hash to use. Default: 0 (`xor_reduction`)

Returns the hash of each row of the input tensor in the given dimension `dim` given by `mode`. If `dim` is a list of dimensions, reduce over all of them.

If `keepdim` is `True`, the output tensor is of the same size as input except in the dimension(s) `dim` where it is of size 1. Otherwise, `dim` is squeezed (see `torch.squeeze()`), resulting in the output tensor having 1 (or `len(dim)`) fewer dimension(s).

`input (Tensor)` – the input tensor.

`dim (int or tuple of ints, optional)` – the dimension or dimensions to reduce. If `None`, all dimensions are reduced.

`keepdim (bool, optional)` – whether the output tensor has `dim` retained or not. Default: `False`.

`mode (int)` – The hash to use. Default: 0 (`xor_reduction`)

